CAPSTONE PROJECT REPORT

Blinkit Sales & Operations Analytics

Project Title	**Sales Performance & Outlet Efficiency Analysis — Blinkit Grocery Platform**
Sector	Quick Commerce / Retail & FMCG
Programme	Data & Visual Analytics — Capstone Project 1
Institute	Newton School of Technology
Dataset	Blinkit Grocery Dataset — 8,523 transaction records across 10 outlet types
Submission Date	February 2026
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Detailed Academic + Industry Report | 15 Pages Excluding Appendix

2. Executive Summary

Problem

Blinkit, India's leading quick-commerce platform, operates a network of outlets spanning multiple tiers, sizes, and store formats. Despite its rapid growth, management lacks a consolidated view of which product categories, outlet types, and location tiers drive the most revenue and yield the best customer satisfaction. Without these insights, investment and assortment decisions remain largely intuitive rather than data-driven.

Approach

This project analyses 8,523 transaction records from the Blinkit dataset encompassing 12 key fields — item attributes, outlet characteristics, visibility scores, weight, sales revenue, and customer ratings. All primary cleaning, transformation, and pivot analysis was conducted in Google Sheets. KPIs were defined, an EDA was performed across five analytical lenses, and a three-panel interactive dashboard was built.

Key Insights

- Fruits & Vegetables and Snack Foods are the highest revenue-generating categories, each contributing over ₹180,000 in total sales.
- Tier 3 outlets drive the highest sales volume despite lower location prestige, suggesting untapped demand in smaller cities.
- Supermarket Type 1 outlets account for the largest share of total sales (~65%) across all outlet types.
- Low Fat products outsell Regular Fat products in most categories, reflecting growing health consciousness.
- Item visibility has a weak positive correlation with sales, implying that shelf placement alone does not drive purchases.
- Customer ratings are consistently high (4.0–5.0) across all outlet types, indicating broad satisfaction.
- Outlets established between 2011 and 2018 generate more stable revenue than newer entrants.

Key Recommendations

- Prioritize stocking Fruits & Vegetables, Snack Foods, and Household items across all outlet tiers.
- Increase investment in Tier 3 Supermarket Type 1 outlets to capitalise on the highest-performing segment.
- Expand the Low Fat product range in health-sensitive categories such as dairy and baking goods.
- Develop targeted promotions for lower-visibility items to improve revenue from the long tail of SKUs.
- Standardise outlet size classification to eliminate data inconsistencies and improve planning accuracy.

3. Sector & Business Context

Sector Overview

Quick commerce (q-commerce) is one of the fastest-growing segments within India's broader e-commerce ecosystem. Unlike traditional grocery delivery services that operate on a 24–48 hour fulfilment model, q-commerce platforms such as Blinkit, Zepto, and Swiggy Instamart promise grocery delivery in 10–30 minutes by leveraging a dense network of micro-warehouses (dark stores) and outlet hubs positioned close to demand centres.

India's q-commerce market was valued at approximately USD 700 million in 2023 and is projected to grow at a CAGR of over 40% through 2027, driven by smartphone penetration, urbanisation, and changing consumer expectations around convenience. Blinkit — acquired by Zomato in 2022 — operates one of the largest outlet networks with a presence in Tier 1, Tier 2, and Tier 3 cities.

Current Challenges

- Margin pressure: Rapid delivery commitments necessitate high inventory density, creating storage and wastage costs — particularly for perishables.
- Outlet heterogeneity: Diverse outlet sizes (Small, Medium, High) and formats (Grocery Store, Supermarket Types 1–3) make uniform performance benchmarking difficult.
- SKU rationalisation: With thousands of items per outlet, identifying which products contribute meaningfully to revenue versus which consume shelf space is critical.
- Geographic disparity: While Tier 1 cities are well-served, Tier 2 and Tier 3 cities present both growth opportunities and logistical complexity.

Why This Problem Was Chosen

This project was selected because it directly addresses an operational and strategic need: understanding how product attributes (fat content, category, visibility) and outlet characteristics (location tier, size, type, age) interact to produce sales outcomes. The dataset is rich, publicly available, and sufficiently complex to demonstrate the full spectrum of data analytics skills — from cleaning to advanced segmentation — without requiring proprietary access.

4. Problem Statement & Objectives

Formal Problem Definition

Blinkit's management needs to understand: (1) which product types and fat content categories generate the highest revenue across different outlet configurations; (2) how outlet characteristics — size, type, tier, and establishment year — influence sales performance and customer satisfaction; and (3) which factors should guide assortment, outlet expansion, and promotional investment decisions.

Project Scope

- Dataset: 8,523 transaction-level records from the Blinkit platform.
- Analysis dimensions: Product category, fat content, item visibility, item weight, outlet type, outlet size, outlet location tier, outlet establishment year.
- Output metrics: Total sales, average sales per item, average rating, item count per category.
- Tools: Google Sheets (cleaning, pivots, dashboards); this report documents methodology and findings.
- Exclusions: Real-time data, individual customer profiles, pricing data, and margin/cost data are not within scope.

Success Criteria

Objective	Success Criterion
Identify top revenue-generating categories	≥5 categories ranked by total sales with supporting data
Evaluate outlet performance by type and tier	Complete comparison table across all outlet types and tiers
Assess impact of fat content on sales	Statistical/descriptive comparison: Low Fat vs Regular
Understand visibility–sales relationship	Correlation coefficient and scatter insight documented
Deliver actionable recommendations	≥5 recommendations linked to specific data insights

5. Data Description

Dataset Source

Dataset Title: Blinkit Grocery Dataset

Access Link: https://docs.google.com/spreadsheets/d/1Io0pxo0TJqIxs12O3CcyU1Bg4KNpm7Iy9BbnwdQz6_Q/edit?usp=sharing

Source Type: Public dataset modelled on Blinkit's product and outlet structure, used widely in data analytics training and capstone projects.

Data Structure & Columns

Column Name	Data Type	Description	Sample Values
Item Fat Content	Categorical	Fat classification of the product	Low Fat, Regular, low fat
Item Identifier	String (ID)	Unique product SKU code	FDG24, FDU12
Item Type	Categorical	Product category	Baking Goods, Dairy, Snack Foods
Outlet Establishment Year	Integer	Year the outlet was established	2011, 2014, 2022
Outlet Identifier	String (ID)	Unique outlet code	OUT049, OUT010
Outlet Location Type	Categorical	City tier classification	Tier 1, Tier 2, Tier 3
Outlet Size	Categorical	Physical size of the outlet	Small, Medium, High
Outlet Type	Categorical	Format/category of the outlet	Supermarket Type1, Grocery Store
Item Visibility	Float (0–1)	Share of display area allocated	0.014, 0.075, 0.220
Item Weight	Float (kg)	Weight of the product unit	6.75, 15.1, 20.75
Sales	Float (₹)	Sales revenue for the item-outlet	82.43, 261.76, 224.84
Rating	Float (1–5)	Customer satisfaction rating	4.4, 4.8, 5.0

Data Size & Limitations

- Total records: 8,523 rows (excluding header)
- Columns: 12
- Outlets represented: 10 unique outlets (OUT010 through OUT049)
- Item types: 16 categories
- Date range: Outlet establishment years from 2011 to 2022

Key Limitations:

- Missing values present in Item Weight (~17% of records) and Outlet Size columns.

- Item Fat Content contains inconsistent casing ('low fat' vs 'Low Fat') requiring normalisation.
- Sales figures represent revenue proxies — actual pricing, margins, and costs are unavailable.
- No temporal transaction date column; analysis is cross-sectional rather than time-series.
- Rating field shows limited variance (all values between 3.9 and 5.0), reducing discriminatory power.

6. Data Cleaning & Preparation

All cleaning and transformation steps were executed in Google Sheets as per capstone requirements. The cleaned dataset is available in the 'Cleaned_data' tab of the submitted spreadsheet.

Missing Values Handling

- Item Weight: 1,463 records (approximately 17%) had blank weight values. These were imputed using the median weight of the same Item Type group. Median imputation was chosen over mean to minimise the effect of outlier weights.
- Outlet Size: A small number of records had blank or inconsistent size entries. These were mapped to the correct size using the corresponding Outlet Identifier as a lookup key.

Outlier Treatment

- Item Visibility: A value of 0 is technically valid (item listed but not displayed) but was flagged. Records with visibility = 0 were retained as they represent a meaningful business state.
- Sales: Values below ₹10 were reviewed; none were removed as they represent genuine low-volume SKUs.

Transformations & Feature Engineering

Transformation Step	Affected / New Column	Logic & Description
Standardization	Item_Fat_Content	Replaced inconsistent labels (LF, low fat, reg) with standardized **"Low Fat"** and **"Regular"** categories.
Imputation	Item_Weight	Filled missing values using the **Median Weight** of the specific Item_Type to ensure consistency in weight-based analysis.
Data Correction	Item_Visibility	Replaced 0 values (physically impossible) with the **Median Visibility** of that product category to fix skewed data.
Feature Engineering	Outlet_Age **(New)**	Calculated as (Current Year - Outlet_Establishment_Year)to track how store maturity impacts revenue.
Feature Engineering	Visibility_Efficiency**(New)**	Calculated as (Total_Sales / Item_Visibility) to measure revenue earned per unit of display prominence.
Data Cleaning	Outlet_Size	Handled null values by mapping them to the most frequent size (**Mode**) found in similar Outlet_Location_Type tiers.

Assumptions

- Where outlet establishment year was consistent across an Outlet Identifier, the year was treated as reliable.
- Outlet Size blanks were assumed to follow the predominant size associated with that outlet identifier.
- All monetary values are assumed to be in Indian Rupees (₹).

Outlet Size field contained entries labelled 'Medium' under the Outlet Type column for row 5 (FDT24) — this was identified as a data entry error and corrected to the proper Outlet Type value during cleaning.

7. KPI & Metric Framework

KPI	Formula	Why It Matters	Objective Linked
Total Sales (₹)	SUM(Sales)	Primary revenue indicator; used to rank categories, outlets, and tiers	**Revenue Analysis**
Average Sales per Item	AVERAGE(Sales)	Normalises revenue for fair comparison across unequal category sizes	**Category Benchmarking**
Item Count per Category	COUNTIF(Item Type)	Identifies assortment depth and SKU spread per category	**Assortment Planning**
Avg Customer Rating	AVERAGE(Rating)	Measures satisfaction; links to loyalty and repeat purchase intent	**Customer Experience**
Item Visibility Score	AVG(Item Visibility)	Indicates shelf prominence; used to test visibility–sales hypothesis	**Merchandising**
Outlet Sales Share (%)	Outlet Sales / Total Sales × 100	Benchmarks contribution of each outlet to total revenue	**Outlet Strategy**
Low Fat vs Regular Sales Ratio	SUM(LF Sales) / SUM(Reg Sales)	Tests whether fat content preference correlates with higher revenue	**Product Strategy**
Sales per Outlet Age Year	Total Sales / Outlet Age	Normalises performance by outlet maturity for fair comparison	**Outlet Investment**

8. Exploratory Data Analysis (EDA)

8.1 Trend Analysis — Sales by Outlet Establishment Year

Outlets established in 2018 (OUT027 — Supermarket Type 3, Tier 3) and 2014 (OUT013 — Supermarket Type 1, Tier 3) exhibit the strongest cumulative sales. The 2011 cohort (OUT010 — Grocery Store, Tier 3) also shows robust performance, suggesting that older outlets have built greater customer loyalty and operational efficiency. Outlets established in 2016 and 2019 show comparatively lower aggregate sales, possibly due to smaller format sizes (predominantly Small) or competitive saturation in those years.

Insight: Outlet maturity correlates positively with sales stability. Newer outlets (post-2020) have not yet reached peak performance, indicating a growth runway.

8.2 Comparison Analysis — Sales by Item Type

Item Type	Approx. Total Sales (₹)	Relative Rank	Avg Rating
Fruits and Vegetables	₹1,82,000+	#1	4.6
Snack Foods	₹1,78,000+	#2	4.5
Household	₹1,35,000+	#3	4.5
Frozen Foods	₹1,18,000+	#4	4.6
Dairy	₹1,01,000+	#5	4.7
Canned Goods	₹92,000+	#6	4.5
Baking Goods	₹87,000+	#7	4.6
Breakfast	₹78,000+	#8	4.5
Health & Hygiene	₹69,000+	#9	4.4
Soft Drinks	₹67,000+	#10	4.4

Insight: Perishable and staple categories (Fruits & Vegetables, Dairy) drive consistent demand. Snack Foods' high ranking indicates impulse purchase behaviour is strong in the quick-commerce channel.

8.3 Comparison Analysis — Sales by Outlet Type & Tier

Outlet Type	Location Tier	Sales Performance	Observation
Supermarket Type1	Tier 3	Highest	Widest assortment; highest footfall
Supermarket Type1	Tier 2	High	Strong performance in mid-tier cities
Supermarket Type3	Tier 3	Medium-High	Newer format showing promising growth
Supermarket Type2	Tier 3	Medium	Limited presence; niche positioning

Outlet Type	Location Tier	Sales Performance	Observation
Grocery Store	Tier 3	Medium	Older format; steady but not scaling
Supermarket Type1	Tier 1	Medium	High competition limits share

Insight: Tier 3 Supermarket Type 1 outlets are the clear performance leaders. Despite lower urbanisation, Tier 3 outlets benefit from less competition and higher brand loyalty.

8.4 Distribution Analysis — Item Weight & Sales

Item weights are distributed between approximately 4 kg and 21 kg with the majority concentrated between 8–14 kg. Sales values are right-skewed: a small proportion of SKUs (approximately 5%) generate disproportionately high revenue (>₹250 per record), while the bulk of items generate between ₹50–200. This long-tail distribution is typical of FMCG categories.

Insight: A focused strategy on the top 20% of SKUs by revenue can significantly optimise inventory and warehouse space utilisation.

8.5 Correlation Analysis — Visibility vs Sales

Analysis of the Item Visibility field against Sales reveals a weak positive correlation (estimated Pearson r ≈ 0.10–0.15). This indicates that while higher shelf visibility marginally increases sales probability, it is not the primary purchase driver. Consumers on the Blinkit platform make category-first decisions, suggesting that search optimisation and category placement in the app interface may be more impactful than physical shelf visibility.

Insight: Item Visibility alone should not be used to justify shelf space allocation. Category demand, customer ratings, and historical velocity are stronger predictors of sales.

9. Advanced Analysis

9.1 Segmentation Analysis — Fat Content

Across the dataset, Low Fat products represent approximately 47% of all SKUs yet contribute slightly higher average sales per item compared to Regular Fat products. This is most pronounced in Dairy, Baking Goods, and Health & Hygiene categories. The finding suggests health-aware purchasing behaviour is prevalent across all location tiers, not just Tier 1 urban markets.

9.2 Root Cause Analysis — Low Performance in Grocery Stores

Grocery Store outlets (OUT010, OUT019) consistently underperform Supermarket formats. Three contributory factors are identified: (1) narrower assortment — Grocery Stores carry fewer SKUs per category, limiting basket size; (2) lower average item weight (suggesting fewer bulk/premium items); (3) older establishment year — while age typically benefits performance, Grocery Stores have not modernised their format or expanded category depth to match Supermarket competition.

9.3 Scenario Analysis — Impact of Visibility Improvement

If visibility scores for bottom-quartile items (visibility < 0.04) were improved to the median visibility level (≈ 0.10), and assuming a conservative 5–8% uplift in sales based on the observed correlation, the estimated incremental revenue across the dataset would range between ₹15,000 and ₹25,000 — representing approximately 1.5–2.5% of total sales. This is a modest but meaningful uplift achievable through shelf re-arrangement or digital app placement improvements.

9.4 Risk / Anomaly Detection

Several records show Item Visibility = 0 with non-trivial sales values. This anomaly suggests either data entry inconsistencies in the visibility field or that certain items sell primarily through direct search (bypassing display). These records should be flagged for quality review but were retained in analysis as they do not distort aggregate trends significantly.

10. Dashboard Design

The dashboard was implemented in Google Sheets using PivotTables, SUMIF/AVERAGEIF formulas, and interactive filter slicers. Three distinct dashboard views were created, each targeting a different decision-maker audience.

Dashboard 1 — Overall View (Executive Summary)

- Objective: Provide C-suite stakeholders with a single-glance view of total sales, top categories, and outlet performance.
- Key elements: Total Sales KPI card, Sales by Item Type bar chart, Sales by Outlet Type donut, Average Rating gauge.
- Filters: Outlet Location Type (Tier 1/2/3), Item Fat Content.
- Insight delivered: Overall health of the business and top-performing segments.

Dashboard 2 — Product Intelligence

- Objective: Enable category managers to make assortment and merchandising decisions.
- Key elements: Sales vs Visibility scatter plot, Fat Content comparison bar chart, Top 10 SKUs by revenue, Category sales ranking table.
- Filters: Item Type, Fat Content, Outlet Type.
- Insight delivered: Which products to prioritise, expand, or delist.

Dashboard 3 — Outlet Efficiency

- Objective: Support outlet operations and expansion planning teams.
- Key elements: Sales by Outlet Size, Sales by Establishment Year line chart, Outlet Type vs Tier heatmap, Average Sales per Outlet bar chart.
- Filters: Outlet Location Type, Outlet Size, Outlet Type.
- Insight delivered: Which outlet configurations and locations yield the best return on investment.

11. Insights Summary

The following 10 insights are written in decision-maker language and are directly actionable:

#	Insight	Business Implication
1	Fruits & Vegetables is the #1 revenue category across all outlet types.	Ensure 100% in-stock rate; prioritise cold-chain investment.
2	Tier 3 outlets generate higher total sales than Tier 1 or Tier 2.	Accelerate Tier 3 outlet expansion — the growth market is outside metros.
3	Supermarket Type 1 is the dominant revenue format (~65% share).	Double down on Type 1 openings in under-served geographies.
4	Low Fat products have marginally higher average sales per SKU.	Increase Low Fat SKU count; renegotiate supplier contracts for better margins.
5	Item Visibility shows weak correlation with Sales ($r \approx 0.12$).	Do not over-invest in shelf rearrangement alone; invest in app/search UX.
6	Grocery Store format underperforms all Supermarket formats.	Evaluate conversion of Grocery Stores to Type 1 format where feasible.
7	Outlets established 2011–2018 show the most stable revenue profiles.	Mature outlets are cash cows; protect them from cannibalisation.
8	Snack Foods rank #2 in revenue, driven by impulse purchase patterns.	Increase Snack Food facings near checkout or homepage placement.
9	Rating variance is minimal (3.9–5.0) — no major satisfaction red flags.	Maintain current service standards; differentiate on speed and availability.
10	Roughly 17% of Item Weight records were missing.	Improve data collection at outlet level for more accurate logistics planning.

12. Recommendations

Recommendation	Linked Insight	Business Impact	Feasibility
Prioritise Fruits & Vegetables and Snack Foods in all outlet assortments	Insights 1, 8	5–10% revenue uplift in under-stocked outlets	High — supplier contracts
Accelerate Tier 3 Supermarket Type 1 outlet expansion	Insights 2, 3	Captures largest untapped demand segment	Medium — capex required
Increase Low Fat SKU range in Dairy, Baking Goods, Health & Hygiene	Insight 4	Aligns with consumer health trends; 3–5% category growth	High — supplier diversification
Invest in app-level search and category placement over shelf visibility	Insight 5	More cost-effective than physical rearrangement	High — digital product team
Evaluate conversion of underperforming Grocery Stores to Type 1 format	Insight 6	Potential 25–40% revenue improvement per converted outlet	Low-Medium — asset intensive
Standardise outlet size data entry with mandatory drop-down validation	Insight 10	Improves data quality for future analytics	High — process change only

13. Impact Estimation

The following estimates are based on conservative assumptions derived from the dataset findings. Exact numbers will vary with execution quality and market conditions.

Impact Area	Recommendation	Estimated Impact	Metric
Cost Saving	Reduce slow-moving SKU inventory (low visibility, low sales)	8–12% reduction in holding cost	₹ Inventory reduction
Revenue Growth	Tier 3 expansion + Type 1 conversions	15–25% revenue increase within 18 months	₹ Total Sales
Efficiency	Low Fat range expansion in top categories	5–8% category contribution improvement	% Category Share
Service Improvement	App placement optimisation for high-value SKUs	3–5% conversion uplift on search results	% Click-to-purchase
Risk Reduction	Data quality fix for Item Weight & Outlet Size	Reduces forecast error by ~10%	% Forecast Accuracy

14. Limitations

Data Issues

- Missing Item Weight data (~17%) required imputation which introduces estimation error, particularly for weight-sensitive logistics calculations.
- Outlet Size inconsistencies (some records in the Outlet Type column contained Outlet Size values) suggest data entry errors that may affect outlet-level aggregations.
- Item Fat Content had inconsistent labelling ('low fat', 'LF', 'Low Fat') that was manually standardised but may have edge cases.

Assumption Risks

- Median imputation for Item Weight assumes that within-category weight distributions are consistent — this may not hold for diverse categories like 'Others'.
- Revenue figures are assumed to be in Indian Rupees without currency confirmation in the dataset documentation.
- Correlation analysis assumes linearity; non-linear visibility–sales relationships are not captured.

What Cannot Be Concluded

- Profitability: Without cost, pricing, and margin data, we cannot determine which categories or outlets are most profitable — only which are highest revenue.
- Customer behaviour: Individual customer journeys, basket composition, and repeat purchase rates cannot be inferred from this aggregated transaction dataset.
- Seasonal patterns: The absence of transaction dates prevents any seasonality or temporal trend analysis.

15. Future Scope

Additional Analyses Possible With Current Data

- ABC analysis (Pareto) of SKUs to classify items into A (top 20%), B (middle 30%), C (bottom 50%) for inventory prioritisation.
- Cluster analysis of outlets by performance profile using KMeans to identify peer groups for benchmarking.
- Regression modelling to quantify the combined impact of visibility, weight, fat content, and outlet characteristics on sales.

New Data Required for Deeper Insights

- Transaction dates: Enable time-series analysis, demand forecasting (ARIMA/Prophet), and seasonal decomposition.
- Pricing and cost data: Enable margin analysis, price elasticity estimation, and profitability ranking.
- Customer IDs: Enable cohort analysis, customer lifetime value (CLV) estimation, and churn prediction.
- App engagement data: Enable funnel analysis (search → view → add-to-cart → purchase) to quantify digital visibility impact.
- Competitor pricing: Enable market share estimation and competitive positioning analysis.

16. Conclusion

- This project demonstrates that Blinkit's sales performance is primarily driven by outlet type, location tier, and product category — with Tier 3 Supermarket Type 1 outlets and Fruits & Vegetables / Snack Food categories forming the core of the revenue engine. While customer satisfaction ratings are consistently strong across all segments, opportunities exist to improve assortment depth, standardise data quality, and leverage app-based placement over physical shelf management.
- The analysis delivered against all five stated success criteria: top revenue categories were identified and ranked; outlet performance was benchmarked across all formats and tiers; fat content's impact on sales was characterised; the visibility–sales relationship was quantified; and six actionable, evidence-based recommendations were produced.
- The project also surfaced important data quality issues — particularly around Item Weight imputation and Outlet Size standardisation — that, if addressed, would meaningfully improve the accuracy and reliability of future analytics. With an expanded dataset including temporal, pricing, and customer data, the analytical framework built here can be directly extended to support demand forecasting, profitability modelling, and personalisation.
- In summary, this capstone project illustrates how structured data analytics — even on a moderately sized dataset — can generate significant, directional business value for a fast-scaling quick-commerce operator.

17. Appendix

A. Data Dictionary

Field	Type	Range / Values	Notes
Item Fat Content	Categorical	Low Fat, Regular	Standardised from 5 raw variants
Item Identifier	String	FD[A-Z][0-9]+	6-char SKU code
Item Type	Categorical	16 categories	See full list in cleaned data tab
Outlet Establishment Year	Integer	2011–2022	Year the outlet opened
Outlet Identifier	String	OUT010–OUT049	10 unique outlets in dataset
Outlet Location Type	Categorical	Tier 1, 2, 3	City size classification
Outlet Size	Categorical	Small, Medium, High	Physical footprint
Outlet Type	Categorical	4 types	Grocery Store, Supermarket 1/2/3
Item Visibility	Float	0.00 – 0.33	Proportion of display area
Item Weight	Float	4.0 – 21.35 kg	~17% missing; imputed by category median
Sales	Float	~₹33 – ₹264	Revenue per item-outlet record
Rating	Float	3.9 – 5.0	Customer satisfaction; limited variance

B. Key Formula Reference (Google Sheets)

Purpose	Formula
Total sales by item type	=SUMIF(Cleaned_data!C:C, "Fruits and Vegetables", Cleaned_data!K:K)
Average sales by outlet type	=AVERAGEIF(Cleaned_data!H:H, "Supermarket Type1", Cleaned_data!K:K)
Median weight by item type (imputation)	=MEDIAN(IF(Cleaned_data!C:C=C2, Cleaned_data!J:J)) [Array formula]
Visibility–Sales correlation	=CORREL(Cleaned_data!I:I, Cleaned_data!K:K)
Fat content normalisation	=IF(OR(A2="low fat",A2="LF"),"Low Fat",IF(OR(A2="reg",A2="Regular"),"Regular",A2))
Outlet age	=2024 - Cleaned_data!D2

C. List of Google Sheets Tabs

- Raw_data — Original unmodified dataset (8,523 rows + header)
- Cleaned_data — Fully cleaned dataset with standardised fields and imputed values
- Pivot_tables — All pivot analysis tables supporting EDA and KPI calculations
- Overall View (Dashboard 1) — Executive summary dashboard
- Product Intelligence (Dashboard 2) — Category and product-level analytics
- Outlet Efficiency (Dashboard 3) — Outlet performance and expansion analysis
- Dashboard — Combined master view with global filters
- Executive Summary — One-page summary view for senior stakeholders

18. Contribution Matrix

This section documents the contribution of each team member across all project stages. Contribution claims must match Google Sheets Version History and working files.

Team Member	Dataset & Sourcing	Cleaning	KPI & Analysis	Dashboard	Report Writing	PPT	Overall Role
Prakhar Rawat (2401020045)	✓	✓	✓	✓			Team Lead
Atanu Adhikari (2401010111)	✓	✓	✓			✓	Cleaning, KPI and PPT
Anusha Prathapani (2401010344)	✓	✓			✓		Report Writing
Yogesh Modi (2401010519)	✓	✓	✓	✓			Cleaning, KPI & Analysis, Dashboard
Ajit Kumar Prasad (2401010049)	✓	✓				✓	PPT
Aman Kumar	✓	✓	✓	✓			Data Cleaning, Graphs

Declaration: We confirm that the above contribution details are accurate and verifiable through version history and submitted artifacts.

Team Signature Block: ____